SET 7: THE LIGEIA MARE HYDROCARBON SUBSURFACE PIPELINE & LIQUEFACTION TERMINAL

DOCUMENT 7A: SECURED CREDIT AGREEMENT

SECURED TERM CREDIT AGREEMENT

Dated as of the Agreement Effective Date (as defined herein)

among

LIGEIA TRANSPORT & LIQUEFACTION LP,
as Borrower

and

JOVIAN-SATURNINE COLONIAL DEVELOPMENT BANK (JSCDB),
as Administrative Agent, Collateral Agent, and Lead Arranger

and

THE FINANCIAL INSTITUTIONS PARTY HERETO FROM TIME TO TIME, as Lenders

---

This SECURED TERM CREDIT AGREEMENT (this "Agreement") is entered into on the Agreement Effective Date by and among Ligeia Transport & Liquefaction LP, as Borrower, the Lenders party hereto, and Jovian-Saturnine Colonial Development Bank, as Administrative Agent. In consideration of the mutual covenants and agreements herein contained, the parties hereto agree as follows:

WHEREAS, the Borrower owns and operates a marine-subsurface pipeline network and cryogenic liquefaction facility transporting and processing liquid methane/ethane directly out of Ligeia Mare on Titan; and

WHEREAS, the Borrower has requested that the Lenders make loans to finance continuous subsurface pipeline structural reinforcement, centrifugal liquefaction plant servicing, and submarine docking platform upgrades; and the Lenders are willing to do so on the terms and subject to the conditions set forth herein.

ARTICLE I: DEFINITIONS AND INTERPRETATION

Section 1.01. Defined Terms. As used in this Agreement, the following terms have the meanings specified below:

- **"Agreement Effective Date"** means the date on which this Agreement has been fully executed and delivered by all parties hereto, as verified by the date inscribed next to the signatures of the authorized officers in the execution block below (such date, the **"Signature Date"**).

- **"Ligeia Pipeline System"** means the physical submarine piping networks, high-pressure booster stations, cryogenic fractional liquefaction plants, and submarine docking gantries situated at Sector 27, Ligeia Mare coordinates (latitude 79.0° N, longitude 248.0° W) on Titan, Saturn.

- **"Pipeline Throughput Target"** means the mandatory minimum operational transport and processing metric of 9,200 metric tons of liquefied natural gas compounds per Titan sidereal day, as calculated under the provisions of Section 4.01(2) herein.

- **"Subsurface Pipeline Integrity Rating"** or **"SPIR"** means the structural, thermodynamic, and pressure stability rating of the submarine-benthic piping lines, as evaluated on a 100-point scale by the Titan Technical Safety Directorate (TTSD).

- **"Sol System Interstellar Accounting Standards"** or **"SSIAS"** means the unified framework of financial accounting principles approved by the Interstellar Financial Board.

- **"TIDOR"** means the Titan Inter-Dome Offered Rate, which is the floating interbank benchmark rate calculated for Titan-denominated credit lines.

- **"Undersea Sediment Escrow"** means the cash escrow reserve fund mandated to be maintained under Section 4.01(3) for benthic sediment containment and localized pressure containment.

Section 1.02. Rules of Interpretation. Capitalized terms defined in the singular shall have the corresponding meanings in the plural, and vice versa. Any references to specific sections, tables, or documents relate directly to this Agreement unless otherwise specified.

ARTICLE II: THE SECURED CREDIT LINES

Section 2.01. The Facility. Subject to the terms and conditions of this Agreement, each Lender agrees to make a single advance term loan (each a "Facility Loan") to the Borrower on the **Closing Date** (which is identical to the **Signature Date**) in an aggregate principal amount of **7,200,000,000 Sol Credits (SC)**.

Section 2.02. Interest Rate Adjustments.

1. The Facility Loans shall accrue interest daily at a floating rate per annum equal to **TIDOR plus 7.25%**.
2. Should the **Subsurface Pipeline Integrity Rating (SPIR)** fall below 89.0% for any consecutive 5-day cycle, the floating margin shall automatically increase by **3.25% per annum** (the "Pipeline Rupture Risk Premium") until the SPIR is restored to 92.0% or above, as verified by the TTSD.

Section 2.03. Repayment and Amortization. Repayment of the principal shall be executed in equal quarterly installments on the final operational day of each Martian fiscal quarter, starting on the first such date following the **Agreement Effective Date**, in accordance with the amortization schedules appended hereto.

ARTICLE III: REPRESENTATIONS AND WARRANTIES

The Borrower represents and warrants to the Lenders that:

Section 3.01. Marine Hydrocarbon Mining Rights. The Borrower holds exclusive licenses to extract, siphon, and process liquefied hydrocarbons within Sector 27-Ligeia, containing high-purity cryogenic methane and ethane mixtures under the authority of the Titan Regulatory Council (TRC).

Section 3.02. Subsurface Pressure Tolerances. The undersea pipeline casings are constructed using high-tensile titanium-niobium alloys and passive thermal insulation, engineered to withstand external hydrostatic pressure variants up to 250 atmospheres and ambient temperatures down to 90 Kelvin.

ARTICLE IV: COVENANTS

Section 4.01. Affirmative Covenants. So long as any part of the Facility Loan remains outstanding, the Borrower shall:

1. **Pipeline System Maintenance:** Undertake continuous acoustic-resonance structural audits of the undersea pipelines to ensure the SPIR does not fall below the base standard of 90.0%.
2. **Maintain Pipeline Throughput:** Operate the high-pressure booster stations to sustain the **Pipeline Throughput Target** of 9,200 metric tons per Titan day, ensuring supply covenants with orbital transfer transport vessels are satisfied.
3. **Sediment Escrow Account:** Maintain the cash-funded **Undersea Sediment Escrow** with the Administrative Agent, holding a balance of not less than SC 450,000,000 at all times to manage ocean-floor structural anchoring stability.

Section 4.02. Negative Covenants. The Borrower shall not, without the explicit written approval of the Administrative Agent:

1. **System Modifications:** Reroute or depressurize the coolant loops of any active secondary liquefaction columns within the designated coordinates.
2. **Restricted Payments:** Declare or make any dividend distributions to corporate parent entities if the SPIR is below 91.0% or if a Default Event has occurred and is continuing.

ARTICLE V: DEFAULT EVENTS AND REMEDIES

Section 5.01. Default Events. The occurrence of any of the following events shall constitute an Event of Default:

1. **Benthic Containment Rupture:** Any sudden structural piping failure leading to an uncontrollable high-pressure hydrocarbon spill exceeding 5,000 metric tons of liquefied methane within any 24-hour cycle.
2. **Covenant Non-Compliance:** Failure to maintain the required balance in the **Undersea Sediment Escrow** under Section 4.01(3), or failure to meet the **Pipeline Throughput Target** for more than two consecutive quarters.
3. **Bedrock Dissolution:** Any localized sub-surface heating adjacent to pipeline anchor pylons causing structural water-ice bedrock temperatures to exceed 240 Kelvin, threatening the alignment or mechanical structural integrity of the submarine docking platform.

ARTICLE VI: MISCELLANEOUS PROVISIONS

Section 6.01. Governing Law. This Agreement shall be governed by, and construed in accordance with, the Admiralty Law of the Martian Lowlands and the outer-rim industrial treaties governing the Saturnian System.

Section 6.02. Counterparts. This Agreement may be executed in any number of counterparts, each of which when so executed and delivered shall be deemed an original, but all of which together shall constitute one and the same instrument.

LIGEIA TRANSPORT & LIQUEFACTION LP

By: _____
Name: Alistair Crane-Vance
Title: Chief Operating Officer, Pipeline Systems

Date of Execution: **June 26, 2146**

JOVIAN-SATURNINE COLONIAL DEVELOPMENT BANK

By: _____
Name: Diana Sterling
Title: Managing Director, Outer-Rim Assets Division

Date of Execution (the **Signature Date**): **June 26, 2146**